9 Things I Learned About Antibiotic Selection (As a Struggling Intern)

A practical, high-yield framework for how to actually think through antibiotic choice on the floor — host, stability, source, coverage, and the pairing logic nobody fully explains in school.

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BEFORE WE START

Why this guide exists

I struggled with antibiotic selection as an intern. Not because I didn't know the drug names — I had the lists memorized. I struggled because nobody walked me through **the order of questions** that actually gets you to the right answer at the bedside.

This guide is the framework I wish someone had handed me on day one: how to think about the host, the clinical picture, the source of infection, and the coverage logic — in the order that actually matters, before you ever land on a drug name.

It's high-yield for interns, useful for board review, and honestly just the way I think through antibiotics now as an attending.

> **How to use this guide:** Read it once start to finish to absorb the framework. Then keep it as a quick reference — each lesson is self-contained, so you can jump straight to the section you need mid-shift.

01

START HERE · THE HOST

I used to jump straight to the drug. Wrong move — start with the host.

Immunocompetent or immunocompromised? That one question changes everything downstream — how broad I go, how fast, and how aggressively I chase a diagnosis before committing to a regimen.

An immunocompromised host (transplant, neutropenic, on chronic steroids, HIV with low CD4) often gets broad empiric coverage immediately, because waiting for confirmation can be the difference between a treatable infection and a fatal one. A healthy host frequently tolerates a narrower, more targeted approach while cultures are pending.

"*The question I should be asking before anything else: who am I treating, not just what am I treating.*"

02

BACTERICIDAL VS. BACTERIOSTATIC

Nobody told me this was a real clinical decision, not just a vocab word.

I genuinely thought this distinction was a board exam detail — until an attending asked me why I'd picked a bacteriostatic drug for a septic, immunocompromised patient.

Critically ill or immunocompromised patients generally need bactericidal agents — beta-lactams, vancomycin, aminoglycosides — because you need the organism dead, not suppressed.

Bacteriostatic agents — macrolides, clindamycin, linezolid — rely on the host's own immune system to finish the job. That's a real risk when the immune system is already struggling to keep up.

03 CLINICAL STABILITY

"Stable" and "unstable" are not the same conversation.

Is the patient hemodynamically stable or not? Septic or not? This determines urgency, route — IV versus PO — and how broad your empiric coverage needs to be before you have the luxury of narrowing.

A septic, unstable patient does not get the luxury of waiting on cultures. You go broad and aggressive immediately, then de-escalate once sensitivities come back. The cost of being too narrow upfront in an unstable patient is far higher than the cost of broadening unnecessarily.

Stability dictates how aggressive I get, right now — not after I've had time to think about it.

04

FOREIGN BODIES & HARDWARE

The first time I missed a line infection, I learned this the hard way.

Central lines, ports, prosthetic joints, cardiac devices — pacemakers, valves. If hardware is involved, you're probably dealing with biofilm-forming organisms like **Staph epidermidis** or **Staph aureus**.

Here's the part nobody emphasized enough in school: **pulling the infected hardware often matters more than which antibiotic you pick.** Antibiotics alone frequently fail when infected hardware stays in place — the biofilm protects the organism from drug penetration regardless of how appropriate your coverage is.

Source control isn't a backup plan. For device infections, it's often the actual treatment.

05

SITE & DEPTH OF INFECTION

I kept treating every infection the same way until depth taught me otherwise.

Superficial cellulitis, a deep abscess, and bacteremia are three completely different problems — even when the organism is identical. **If it's deep-seated, antibiotics alone aren't enough — it needs source control** (drainage, debridement).

Before ordering anything empirically, run three questions:

Above or below the diaphragm?

Above → think Gram-positive / respiratory flora. Below → think Gram-negative / GI / GU flora.

Is there MRSA or Pseudomonas risk?

Based on prior antibiotics, recent hospitalization, or known colonization history.

Does it need to cross into the CNS?

Not every drug crosses the blood-brain barrier — this changes your options substantially.

06

THE NIGHTMARE BUGS

I had to make myself memorize these instead of looking them up every time.

Three coverage lists worth knowing cold:

MRSA

Vancomycin · Linezolid · Daptomycin · Ceftaroline

Oral: TMP-SMX, Doxycycline

PSEUDOMONAS

Pip-Tazo · Cefepime · Ceftazidime · Carbapenems* · Aztreonam · Cipro / Levo

*Not Ertapenem

ANAEROBES

Below diaphragm → Flagyl
Above diaphragm → Clindamycin

These three lists cover the majority of "I'm worried about resistance" decisions you'll make as an intern.

07 COMBINATION PAIRS

These finally made sense once I stopped memorizing and started asking "what's the gap?"

CAP (Community-Acquired Pneumonia)

Ceftriaxone + Azithromycin. Ceftriaxone nails S. pneumo but completely misses atypicals — Azithromycin sweeps those up. (Or use Levofloxacin alone, which covers both.)

Abdominal Infections

Ceftriaxone + Flagyl. Ceftriaxone handles Gram-negatives but has zero anaerobic coverage — Flagyl fills that hole.

Meningitis

Ceftriaxone + Vancomycin (+ Ampicillin if age > 50). Vancomycin covers resistant S. pneumo. Ampicillin covers Listeria, which Ceftriaxone inherently misses entirely.

08

THE MOST COMMON MISTAKE

This one genuinely confused me for months — "double coverage" ≠ "combination therapy."

I used to think any two antibiotics ordered together meant "double coverage." It doesn't.

True double coverage means two drugs attacking the *same* bug — and it's basically reserved for severe Pseudomonas in critically ill patients (septic shock, ventilator-dependent, neutropenic fever, known MDR risk). The logic: a beta-lactam base (Cefepime, Pip-Tazo, Meropenem) plus a non-beta-lactam companion (an aminoglycoside or fluoroquinolone), so if resistance defeats one drug, the other still works.

Combination therapy is different — it's filling gaps in coverage, not doubling up on the same organism. About **90% of the antibiotic pairs you'll actually see ordered** are this, not true double coverage.

09

CEPHALOSPORIN GENERATIONS

This finally clicked once I saw the pattern instead of memorizing a list.

Each generation trades a little Gram-positive coverage for more Gram-negative reach:

GEN	COVERAGE	CLASSIC EXAMPLE
1st	Gram-positive heavy (Staph / Strep)	*Cefazolin*
2nd	Gram-positive + some Gram-negative	*Cefuroxime*
3rd	Serious Gram-negative, crosses BBB	*Ceftriaxone*
4th	Broad spectrum + Pseudomonas	*Cefepime*
5th	Broad spectrum + MRSA	*Ceftaroline*

5th gen is the outlier — it circles back and adds MRSA coverage, which is what makes it the special exception worth remembering separately.

✦

Save this.
Share it with
your intern.

This is the framework I actually use now — not just for boards, but on the floor, every shift. Hope it saves you some of the confusion it took me a while to work through.

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